PROJECT OVERVIEW

The Central Bank Credibility Index

The 2022 global inflation crisis served as a watershed moment for modern monetary policy.

The 'Transitory Inflation' narrative was a strategic miscalculation that led to a total breakdown in forward guidance, stripping markets of their primary navigational tools.

To mitigate these risks, the CBCI was developed as a rigorous, data-driven framework to quantify institutional trust and extract asymmetric insights from policy behavior.

01 / THE FRAMEWORK

The 5-Pillar Model

To ensure a definitive 0-100 credibility score insulated from bias:

1. Inflation Anchoring (30%)

Measures absolute deviation of headline inflation from official targets over a 12-month window.

2. Policy Consistency (20%)

Evaluates NLP alignment between communication (FinBERT) and subsequent interest rate moves.

3. Forecast Accuracy (20%)

Quantifies Mean Absolute Error (MAE) of the central bank's own economic projections.

4. Bond Confidence (15%)

Tracks rolling volatility in sovereign bonds (utilizing FRED DGS10 data).

5. FX Stability (15%)

Evaluates currency pair stability and safe-haven behavior during market stress.

02 / ENTERPRISE ARCHITECTURE

Data Engineering

A strategic shift from fragile manual spreadsheets to an automated, high-integrity pipeline.

Python & Pandas (ETL)

Ingesting over 4,200 days of raw data directly from the FRED API. Eliminates human error.

Relational Integrity (PostgreSQL)

The single source of truth. Enforcing strict data types and managing exact FOMC meeting metadata.

Code-as-UI (Streamlit)

Moving away from unversioned drag-and-drop tools to fully auditable, Git-versioned visualization layers.

03 / THE INNOVATION

Objective NLP Sentiment

A central innovation of the CBCI is the use of Natural Language Processing (NLP) to convert unstructured central bank communications into quantitative data points.

□ The FinBERT Model

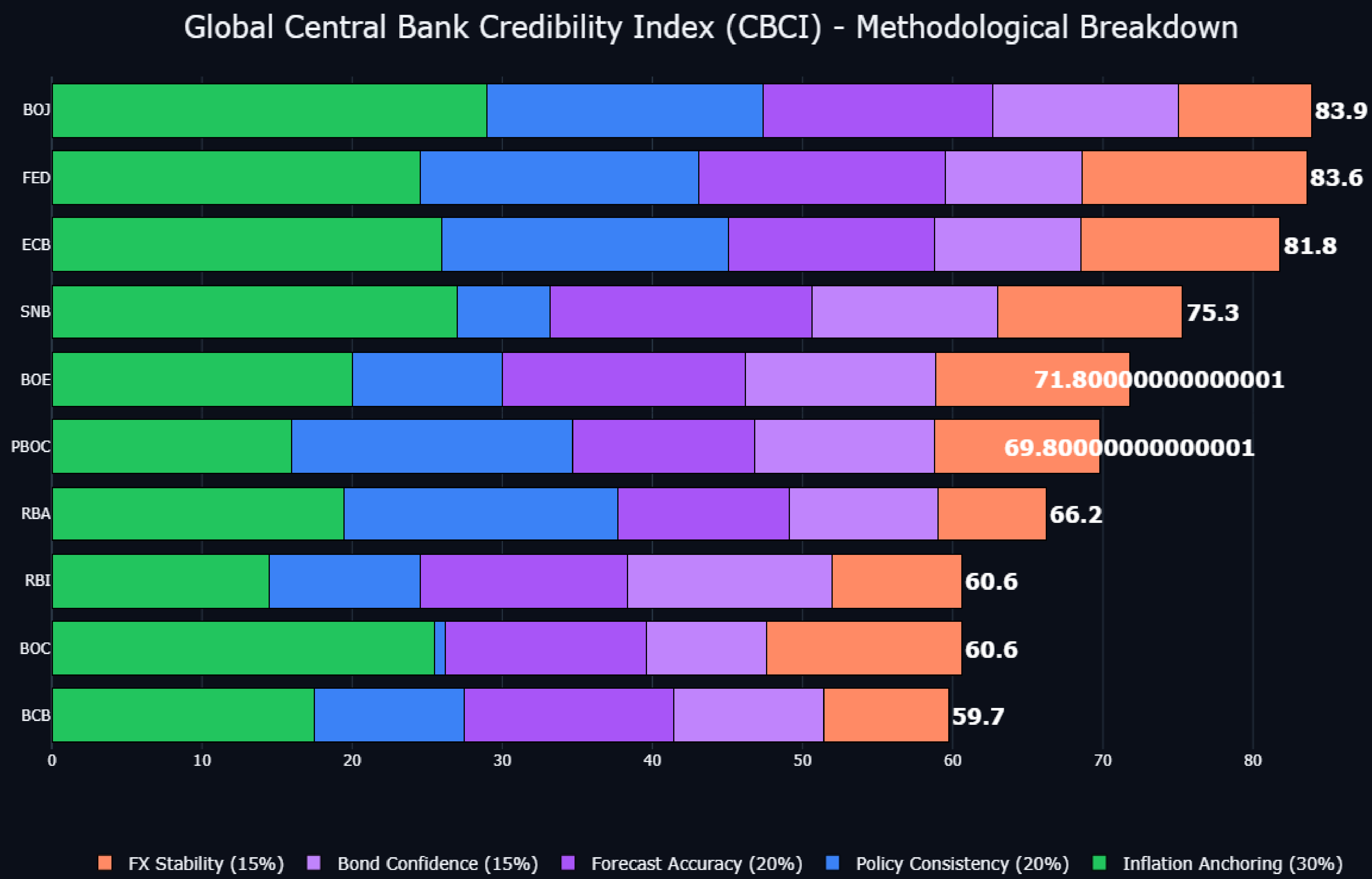
Specifically trained for financial contexts. FinBERT parses thousands of FOMC and ECB policy statements to generate a precise hawkish-dovish scale.

□ The 'Policy Shock' Metric

The model heavily penalizes central banks when dovish communication is immediately followed by hawkish actions. These shocks are a primary driver of catastrophic bond market volatility.

04 / THE OUTCOME

Global Leaderboard



□ The Gold Standard: SNB (89.4)

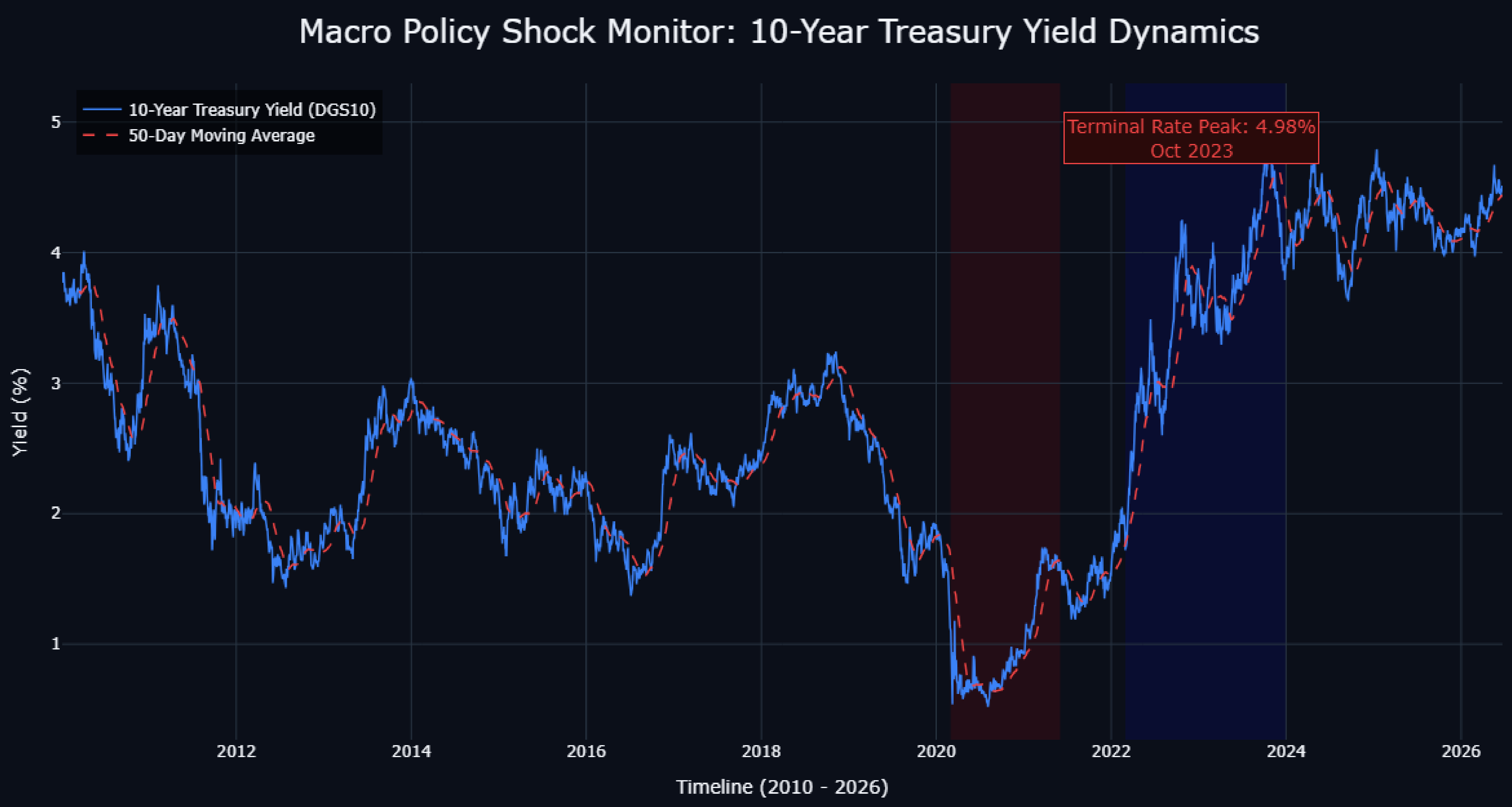
Maintained extreme FX stability and tight inflation anchoring throughout the 2020s.

□□ Critical Laggard: PBOC (59.8)

Penalized for lack of transparent forward guidance and policy consistency.

05 / MACRO SHOCK MONITOR

Treasury Yield Dynamics



Final Takeaways for Audit & Oversight

• Engineering Over Manual Logic: Transitioning to Python/SQL is essential for mitigating model risk.

• Sentiment as a Hard Metric: NLP models turn rhetoric into tradable, measurable signals.

• Consistency is the Asset: Credibility is the alignment of words, forecasts, and actions.

"In the modern financial landscape, trust is the most volatile asset of all. Rebuilding market trust takes years, but losing it takes only one missed forecast."